



Unit-III: Different Algorithms in Machine Learning



3.1

Evaluating Machine Learning algorithms and Model Selection





Introduction

- Why evaluation is important in ML?
- Difference between training performance and real-world performance
- Bias–variance tradeoff

2. Performance Metrics

□ For Classification

- Accuracy, Precision, Recall, F1-score
- Confusion matrix
- ROC curve, AUC

□ For Regression

- Mean Squared Error (MSE), Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)
- R^2 score (coefficient of determination)

□ For Ranking/Recommendation

- Precision@k, Recall@k, MAP, NDCG



3. Validation Techniques

- ❑ Hold-out method (train/test split)
- ❑ k-Fold Cross-Validation
- ❑ Leave-One-Out Cross-Validation (LOOCV)
- ❑ Stratified sampling (for imbalanced datasets)



4. Model Selection Strategies

- ❑ Comparing multiple algorithms
- ❑ Hyperparameter tuning
 - ❑ Grid Search
 - ❑ Random Search
 - ❑ Bayesian Optimization
- ❑ Automated Machine Learning (AutoML)

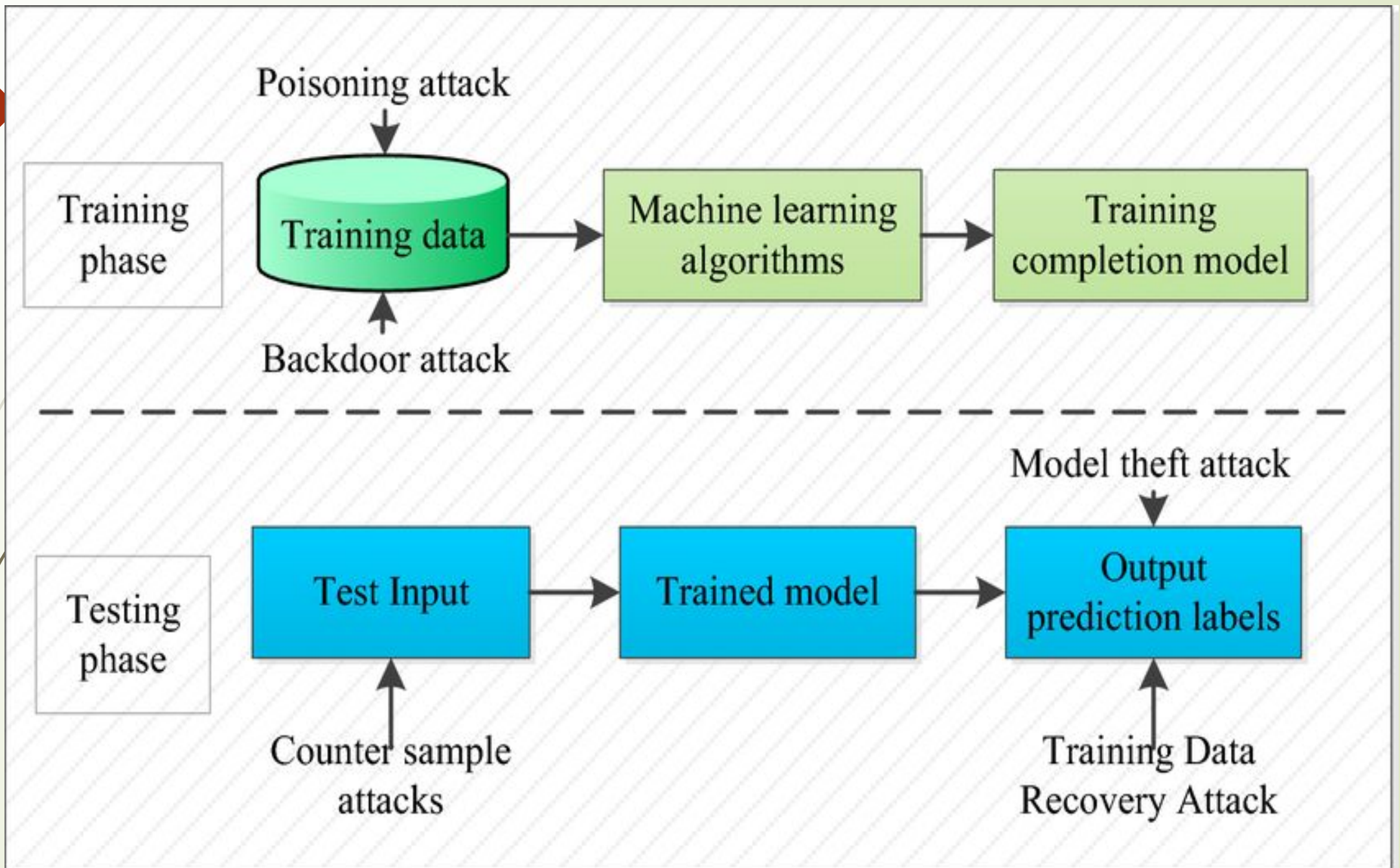
5. Overfitting and Underfitting

- Causes and detection
- Regularization (L1, L2, dropout)
- Early stopping



6. Model Complexity and Generalization

- Bias–variance tradeoff revisited
 - Model capacity vs dataset size
 - Occam's razor principle in ML
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3.2

**Introduction to
Statistical Learning
Theory**

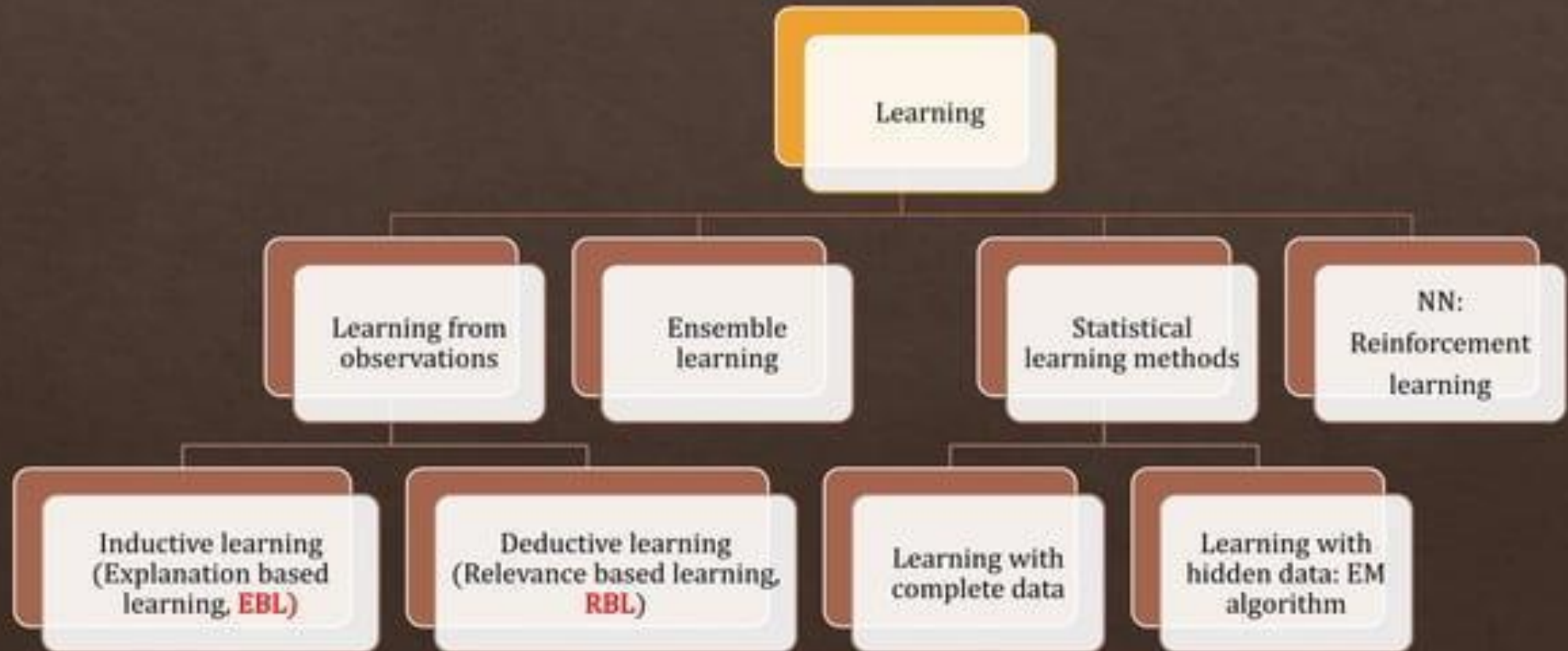
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- Statistical Learning Theory (SLT) is a theoretical framework for understanding **how machines learn from data** and make predictions.
 - It provides the **mathematical foundations** behind many modern machine learning algorithms and helps explain their performance, limitations, and generalization ability.

Statistical Learning Theory

- The goals of learning are **understanding and prediction**.
- Learning falls into many categories i.e. Supervised, unsupervised, online, reinforcement learning.
- The learning problem consists of **inferring the function that maps between the input and the output**, such that the learning function can be used to predict output from future input.



The place of Statistical Learning Methods in LEARNING





□ Learning Problem

- Input: A dataset consisting of feature vectors and corresponding labels.
- Goal: Learn a function (hypothesis) that maps inputs to outputs with minimal error.

□ Hypothesis Space (H)

- The set of all possible models the learning algorithm can choose from.
- Example: In linear regression, the hypothesis space is the set of all linear functions.

• Risk and Loss Functions

- **Loss function:** Measures the error of predictions (e.g., squared error, hinge loss, cross-entropy).
- **Expected Risk (True Risk):** Average loss over the entire data distribution (unknown in practice).
- **Empirical Risk:** Average loss over the training dataset (known).
- **Empirical Risk Minimization (ERM)**
A principle where the learner chooses the hypothesis that minimizes the training error.
- **Problem:** ERM may lead to **overfitting** if the hypothesis is too complex.

• Generalization

- The ability of a model to perform well on **unseen data**.
- Statistical Learning Theory provides tools (like VC dimension, bounds, and
- regularization) to study generalization

• VC Dimension and Capacity Control

- **Vapnik–Chervonenkis (VC) dimension**: A measure of the complexity or capacity of a hypothesis space.
- Models with very high VC dimension can overfit; too low VC dimension may underfit.



•Structural Risk Minimization (SRM)

- An approach that balances **model complexity** and **training error**.
- Introduces **regularization** to avoid overfitting and improve generalization.

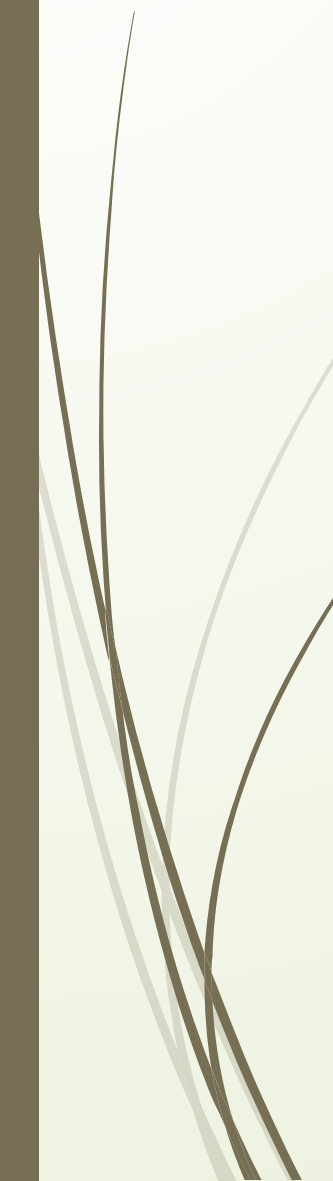
•PAC Learning (Probably Approximately Correct)

- Framework that defines conditions under which a learning algorithm can guarantee
- that the learned hypothesis is close to the true function, with high probability.



3.3

Ensemble Methods (Boosting, Bagging, Random Forests)



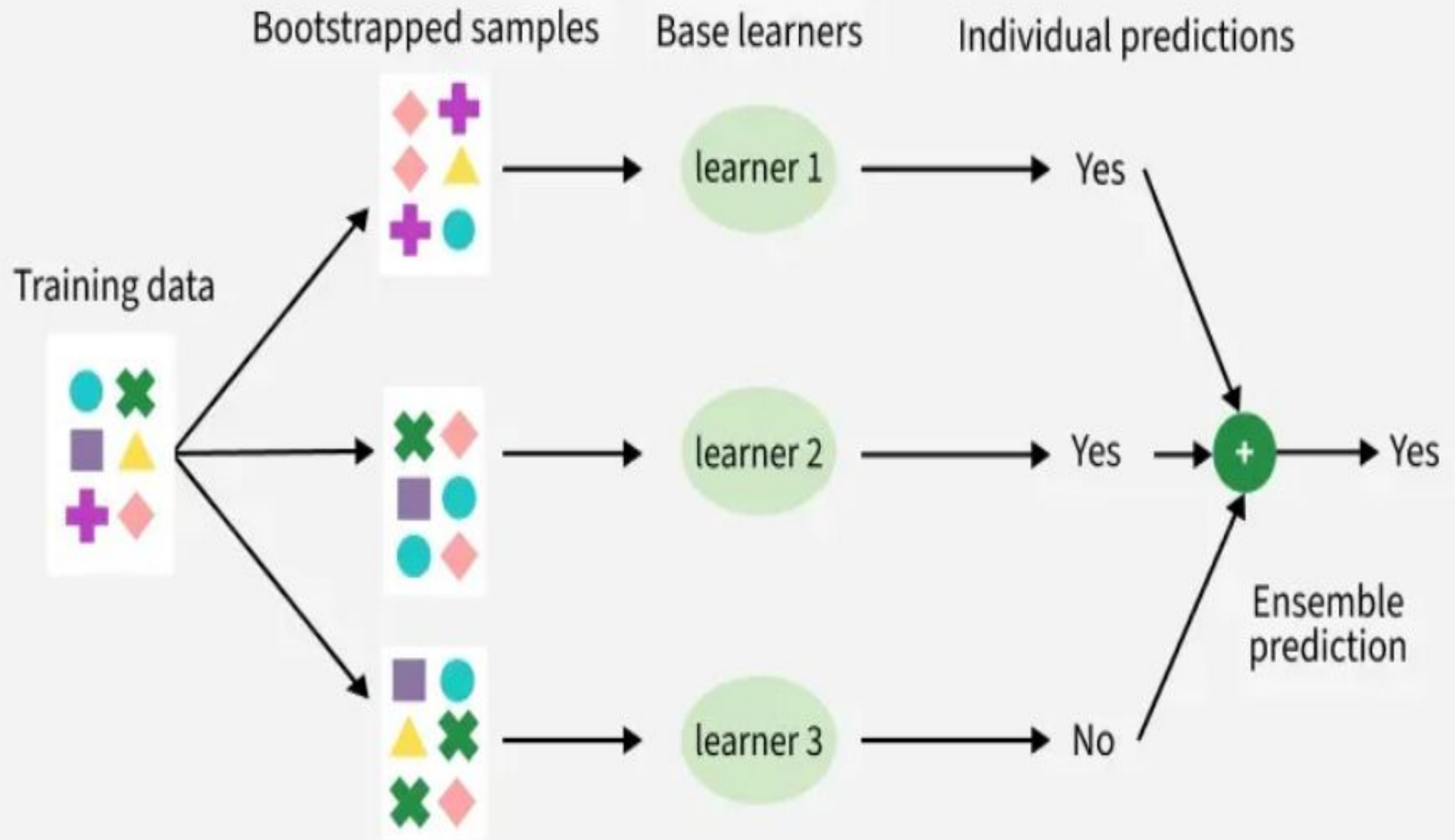
Ensemble learning

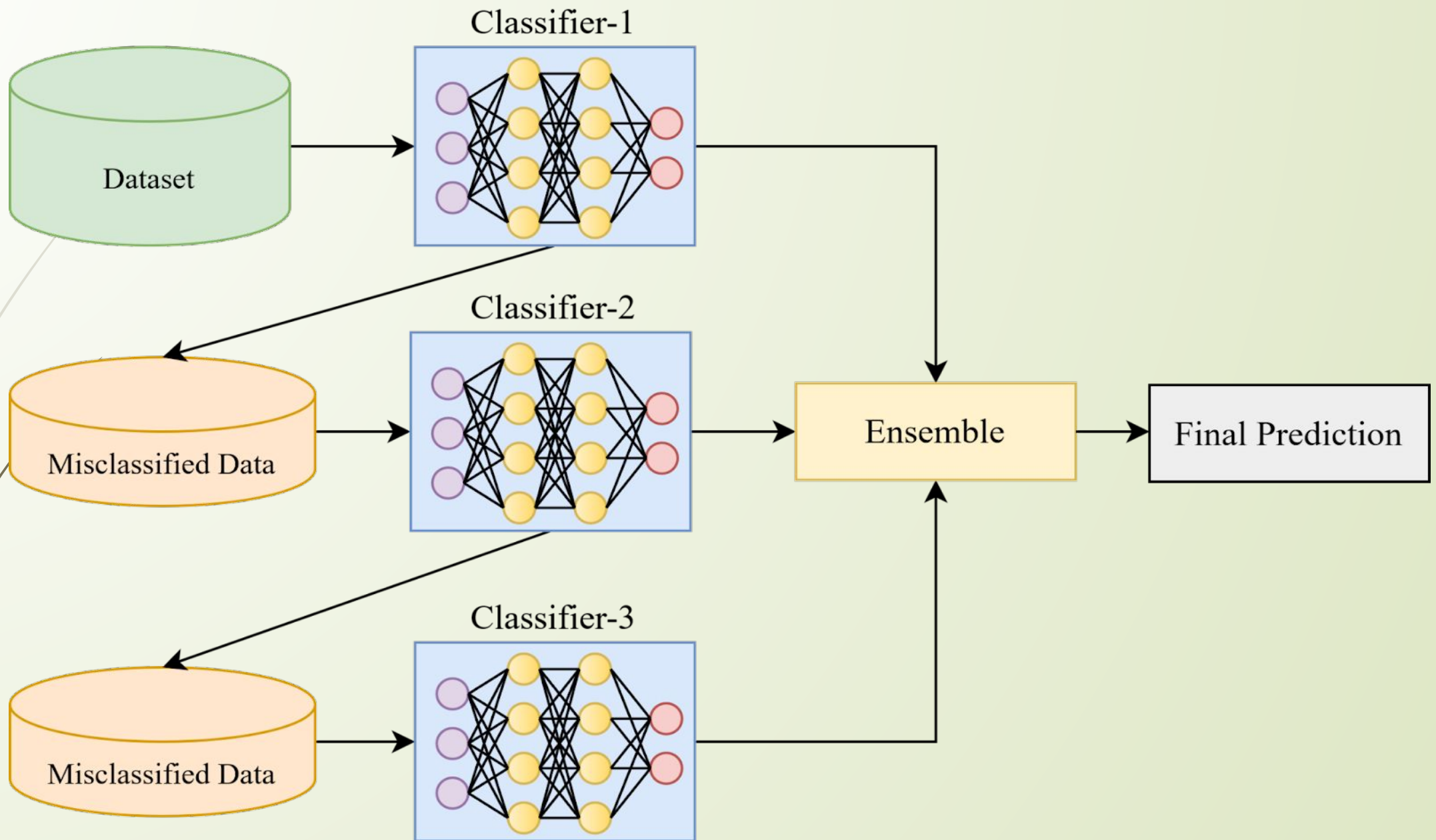
- **Ensemble learning** is a technique in machine learning where **multiple models (often called "weak learners")** are trained and combined to solve the same problem.
- The idea is that **a group of models working together can outperform a single strong model.**

When to Use Ensemble Learning?

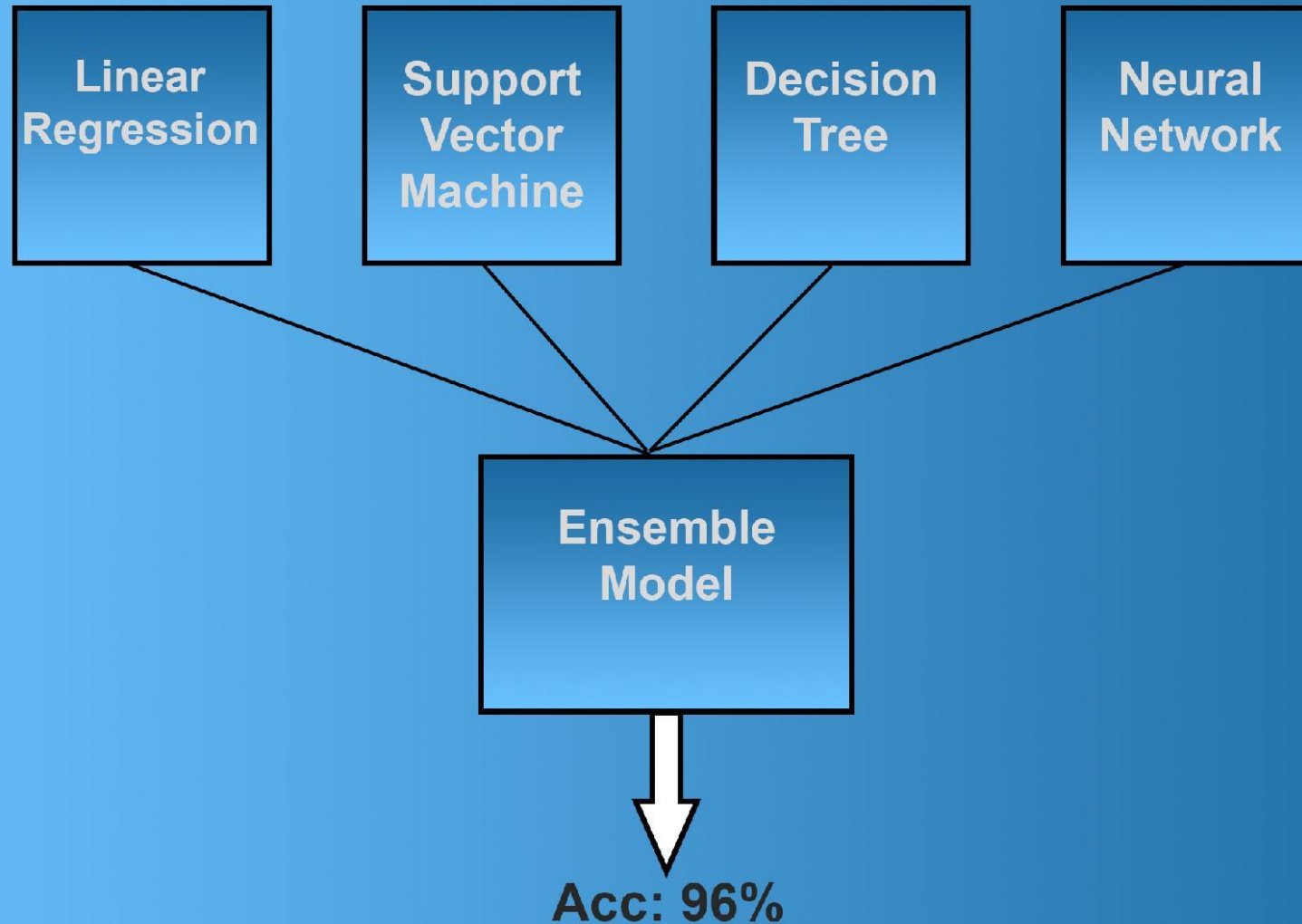
- ❑ You have **high variance or high bias** in your model.
- ❑ Your base models are **diverse and complementary**.
- ❑ You want to **increase model performance** in competitions (e.g., Kaggle).
- ❑ **Real-Life Example**
 - ❑ Imagine trying to guess a movie's rating:
 - ❑ One friend uses past ratings
 - ❑ Another reads online reviews
 - ❑ A third watches the trailer
 - Each has weaknesses alone, but combining their opinions gives a better estimate. That's ensemble learning!

Ensemble learning

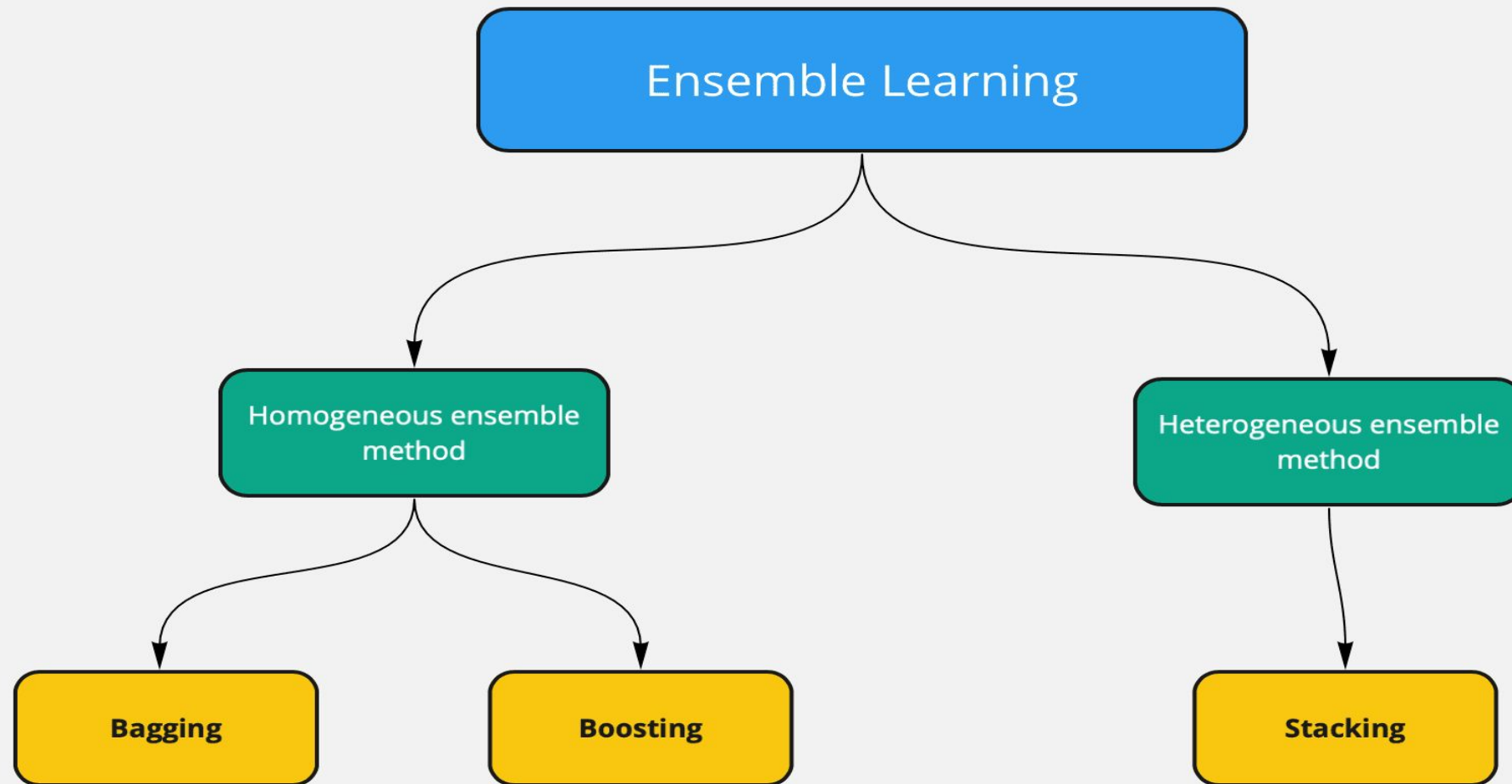




Accuracy: Highest



Types of Ensemble learning

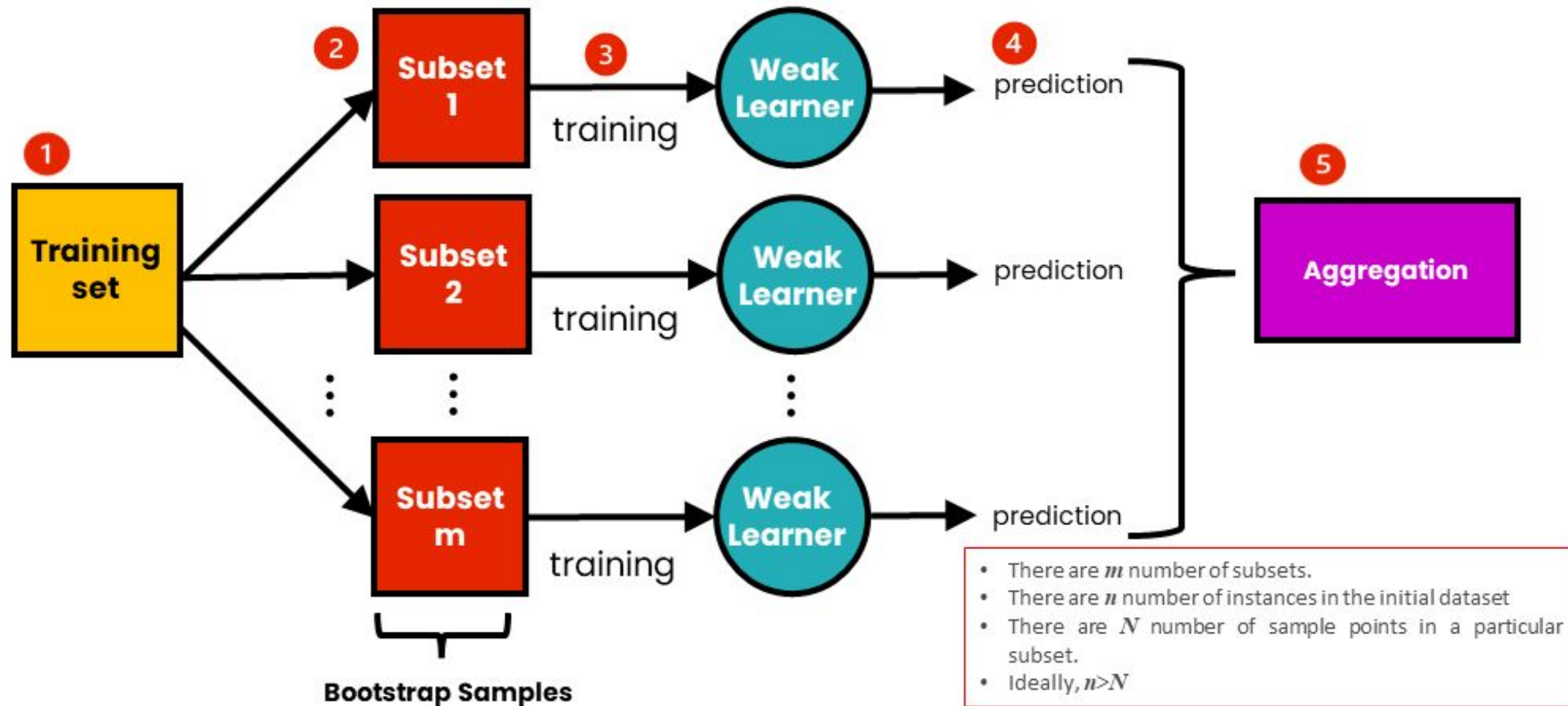


❑ **Bagging (Bootstrap Aggregating):**

- Models are trained independently on different random subsets of the training data.
- ❑ Their results are then combined—usually by averaging (for regression) or voting (for classification).
- ❑ This helps reduce variance and prevents over fitting.
- ❑ **Example Algorithms:**
- ❑ **Random Forest** (uses decision trees + bagging)

How Bagging Learning model works:

The Process of Bagging (Bootstrap Aggregation)

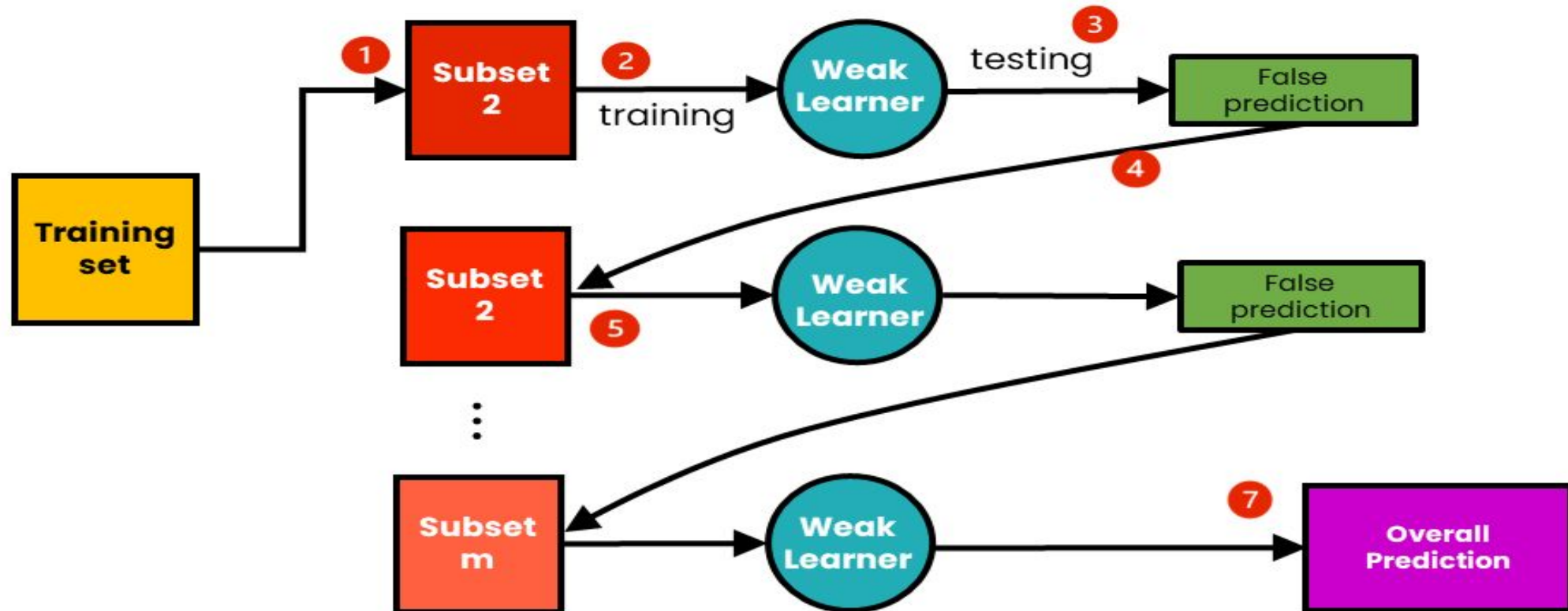


Boosting:

Models are trained one after another. Each new model focuses on fixing the errors made by the **previous ones**.

The final prediction is a weighted combination of all models, which helps **reduce bias and improve accuracy**.

The Process of Boosting





Models are trained **sequentially**, each new model focusing on **correcting the errors** of the previous ones.

□ Final prediction is a **weighted sum** of all models.

□ **Reduces bias and variance**

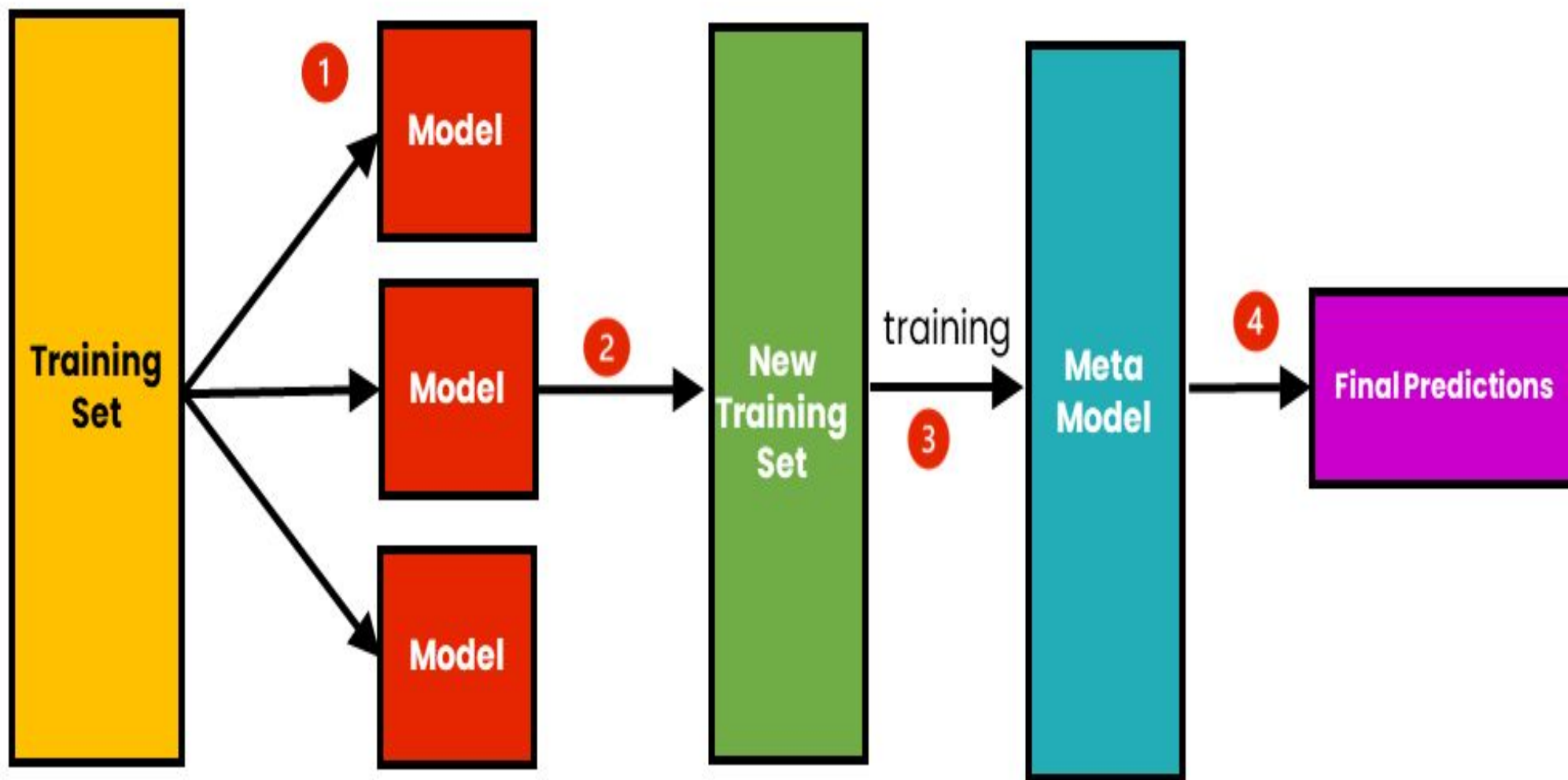
□ **Popular Boosting Algorithms:**

- ✓ AdaBoost (Adaptive Boosting)
- ✓ Gradient Boosting (GBM)
- ✓ XGBoost
- ✓ LightGBM
- ✓ CatBoost

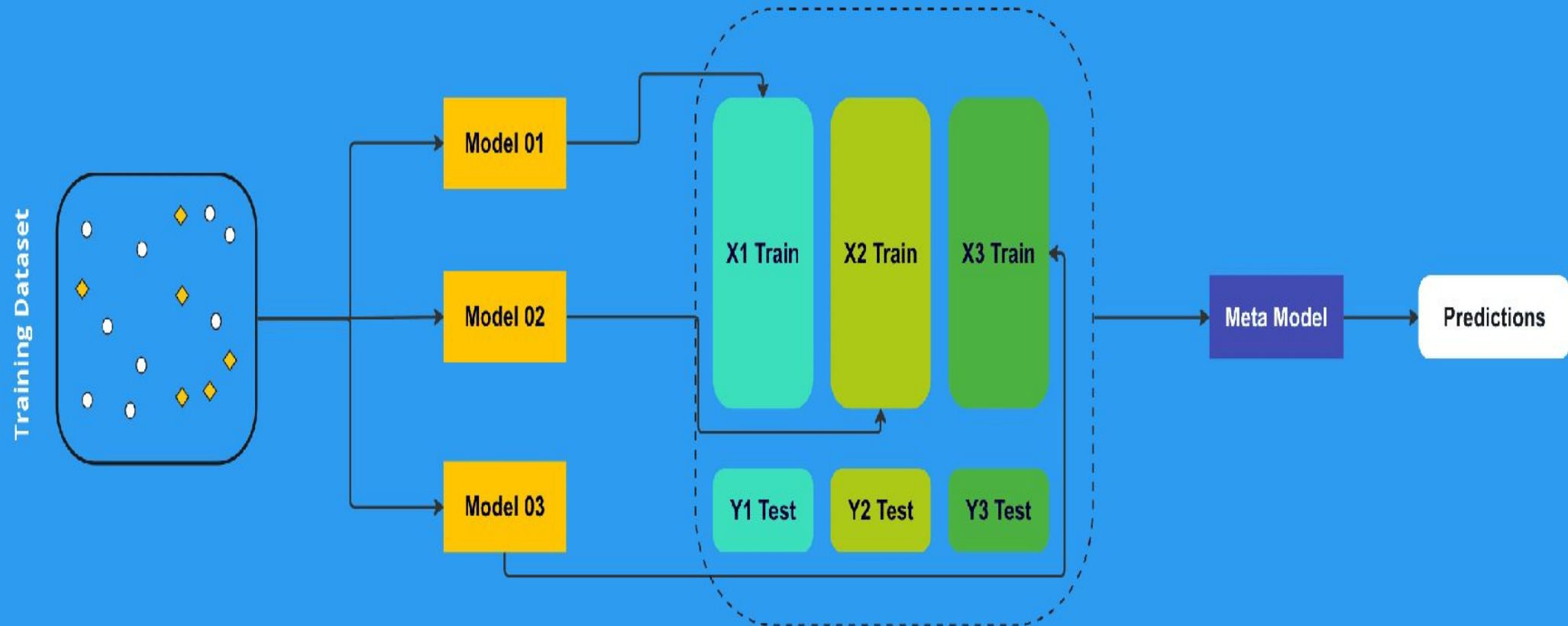
Stacking (Stacked Generalization)

- ❑ Multiple different models (often of different types) are trained, and their predictions are used as inputs to a final model, called a meta-model.
- ❑ The meta-model learns how to best combine the predictions of the base models, aiming for better performance than any individual model.
- ❑ The predictions of base models are fed to a **meta-model** (e.g., logistic regression) that learns how to best combine them.
- ❑ **Leverages strengths of different models**
- ❑ Useful when base learners are diverse.

The Process of Stacking



How Stacking Technique Improves Machine Learning Model's Performance



Model Dataset - - - - -> Level 01 Modeling - - - - -> Level 01 Predictions - - - - -> Level 02 Modeling - - -> Final Predictions

Real-Life Example

- Imagine trying to guess a movie's rating:
- One friend uses past ratings
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- A third watches the trailer

Each has weaknesses alone, but combining their opinions gives a better estimate. That's ensemble learning!